



# Ray Tracing

Ray Tracing is a rendering technique that simulates the physics of light by tracing the path of rays from the camera backward through each pixel and into the scene.

In the real world: Light sources emit millions of rays, rays bounce off objects and some rays eventually hit the human eye.

But this is incredibly inefficient because most rays never reach the human eye.

Instead, Ray Tracing traces rays from the camera:

1. For each pixel, cast one ray into the scene
2. Find the first object it hits
3. Calculate the color at that hit point
4. That's the pixel's color